

Temporal Anti-Aliasing for Single-Photon Reconstruction

CS 566: Introduction to Computer Vision — Project Proposal

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Problem Statement

Single-photon avalanche diode (SPAD) cameras capture high-speed bursts of binary frames, where each pixel records only whether a photon arrived during a given exposure. A single frame carries almost no usable signal; a clean image only emerges from aligning and merging an entire burst (quanta burst photography). This project investigates single-photon image reconstruction under motion, specifically whether history-rejection heuristics from real-time temporal anti-aliasing (TAA) — neighborhood clamping and motion-adaptive blending — improve classical align-and-merge reconstruction, particularly under fast motion and disocclusion.

Motivation

The standard SPAD reconstruction pipeline (estimate motion, align, merge) is structurally identical to temporal anti-aliasing in real-time rendering, yet this connection does not appear to have been explored in prior work. Bringing history-rejection techniques developed for real-time graphics into photon-level reconstruction offers a genuinely cross-domain angle with a clear, testable hypothesis. The project also builds directly on Prof. Gupta's own quanta burst photography research and the Single Photon Challenge benchmark his lab co-organized, giving the team a public dataset with ground truth rather than requiring custom data collection.

Related Work

- Quanta burst photography (Gupta et al.) establishes the align-then-accumulate pipeline for SPAD bursts that serves as our baseline.
- A survey of temporal anti-aliasing techniques (Yang et al., NVIDIA, 2020) catalogs the history-rejection methods — neighborhood clamping, motion-adaptive blend weights — that we adapt to this domain.
- The Single Photon Challenge provides the benchmark dataset and existing leaderboard submissions for comparison.

Approach

1. Implement a baseline: naive frame averaging, followed by hierarchical block alignment and merging (a simplified quanta burst photography pipeline).
2. Implement a TAA-augmented pipeline using the same alignment and merge backbone, adding neighborhood-clamping history rejection, motion-adaptive blend weights, and a per-pixel confidence map.
3. Evaluate both pipelines on held-out Single Photon Challenge training scenes and on synthetic motion sweeps generated with the lab's visionsim simulator.
4. Analyze where and why each method breaks down under fast motion and disocclusion.

This approach draws on course material including image alignment and transformations, RANSAC, optical flow, tracking, and the quanta computational imaging lecture.

Data

The Single Photon Challenge dataset provides 50 simulated training scenes and 5 test scenes (ground truth withheld), each burst ("photon cube") consisting of 1,024 binary frames. The full set is approximately 425 GB; we will work with a 5–10 chunk subset (roughly 8.5 GB per chunk), supplemented with our own synthetic motion sweeps generated via the MIT-licensed visionsim simulator.

Evaluation Plan

We will report PSNR and SSIM on held-out training scenes, and conduct a failure analysis against motion speed using visionsim sweeps to characterize when temporal reconstruction breaks down. The contribution is framed as an interpretable classical method with a rigorous failure analysis, rather than a state-of-the-art claim against neural approaches.

Timeline

Week	Dates	Milestone
1-2	Sep 29 - Oct 12	Data pipeline (subset download, visionsim setup) and baseline (naive averaging + block alignment) working
3-4	Oct 13 - Oct 26	TAA-augmented pipeline implemented (clamping, adaptive blending, confidence map)
5	Oct 27 - Oct 29	Mid-term report: baseline vs. TAA results, difficulties encountered
6-8	Oct 30 - Nov 19	Motion-speed sweeps via visionsim, failure analysis, iteration
9	Nov 20 - Nov 30	Final evaluation, ablations, results polish
10	Dec 1 - Dec 10	Presentation and final webpage

Scope and Feasibility

This project topic was discussed with Prof. Gupta, who confirmed it is a reasonable and worthwhile direction to propose. As is expected at this stage, the specific scope may be refined as implementation progresses and initial results clarify what is achievable within the semester. Prof. Gupta also offered to connect the team with members of his lab for data access or implementation guidance if needed.